AIU Unified Portal vs Open-Source University Finance & HR Repositories

Repository-based gap analysis, Graphify findings, and implementation roadmap

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1. Executive summary

Your AIU Unified Portal is already a broad university portal rather than a small student-management app. It implements admissions, academic directory, semesters, course registration, scheduling, LMS, attendance, results, student services, venue booking, elections, library, canteen, finance subscriptions, ICT helpdesk, chatbot, notifications, reports and audit. The GitHub repository comparison shows that the major missing areas are not basic academic modules; they are enterprise-grade Finance and HR: student ledger, full fee lifecycle, general ledger, procurement/AP, budgets, reconciliation, employee master, recruitment, onboarding/offboarding, staff leave, employee attendance, payroll, deductions/tax, payslips, performance appraisal and HR self-service.

- Finance priority: convert accountant_app from subscription/fine handling into a student-finance subledger, then optionally integrate a full accounting/GL layer.
- HR priority: create a unified hr_app around existing role-detail models, then add employee lifecycle, leave/attendance and payroll in phases.
- Architecture fit: the project already has Profile, role routing, notifications, audit logs, reports, Celery tasks, XLSX exports and payment-gateway patterns, so new HR/Finance modules can be added without replacing the existing portal.
- Open-source reference patterns: Kuali KFS is the best higher-ed finance model; Frappe/ERPNext + Frappe HRMS is the best modular implementation model; Odoo/OCA and OrangeHRM are useful HR/payroll references.

2. Visual comparison

The following charts summarize the 100-repository comparison and the functional coverage gap.

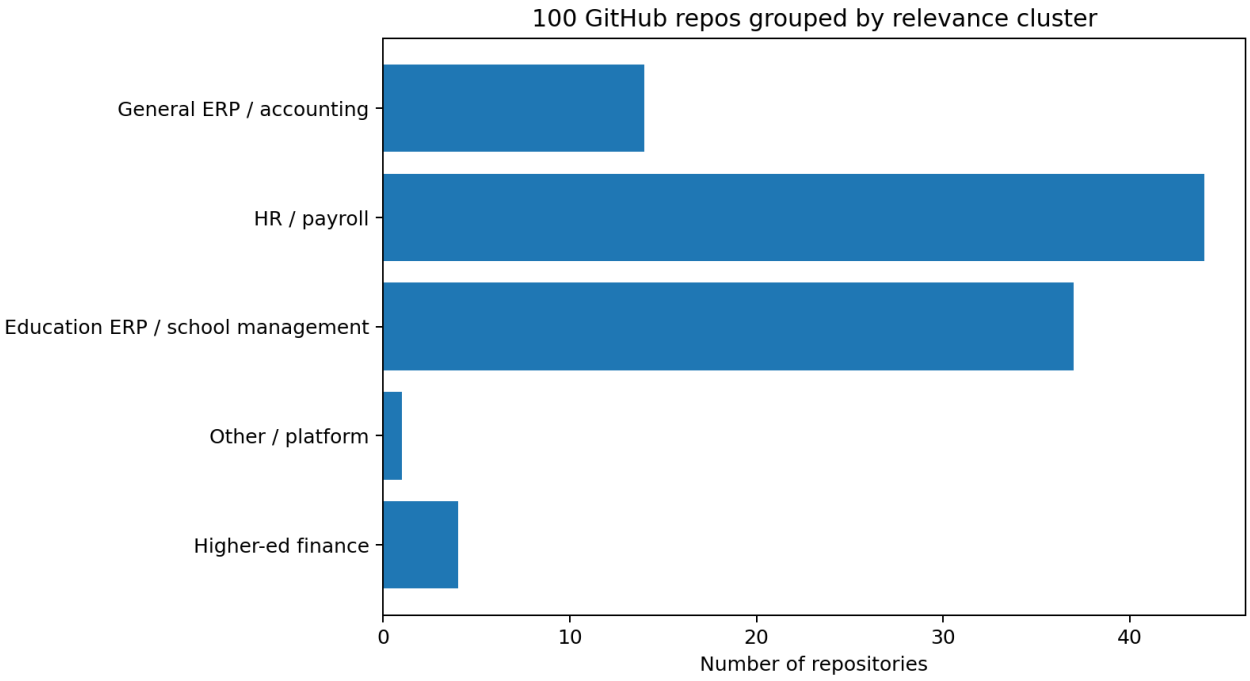


Figure 1. Repository corpus by relevance cluster.

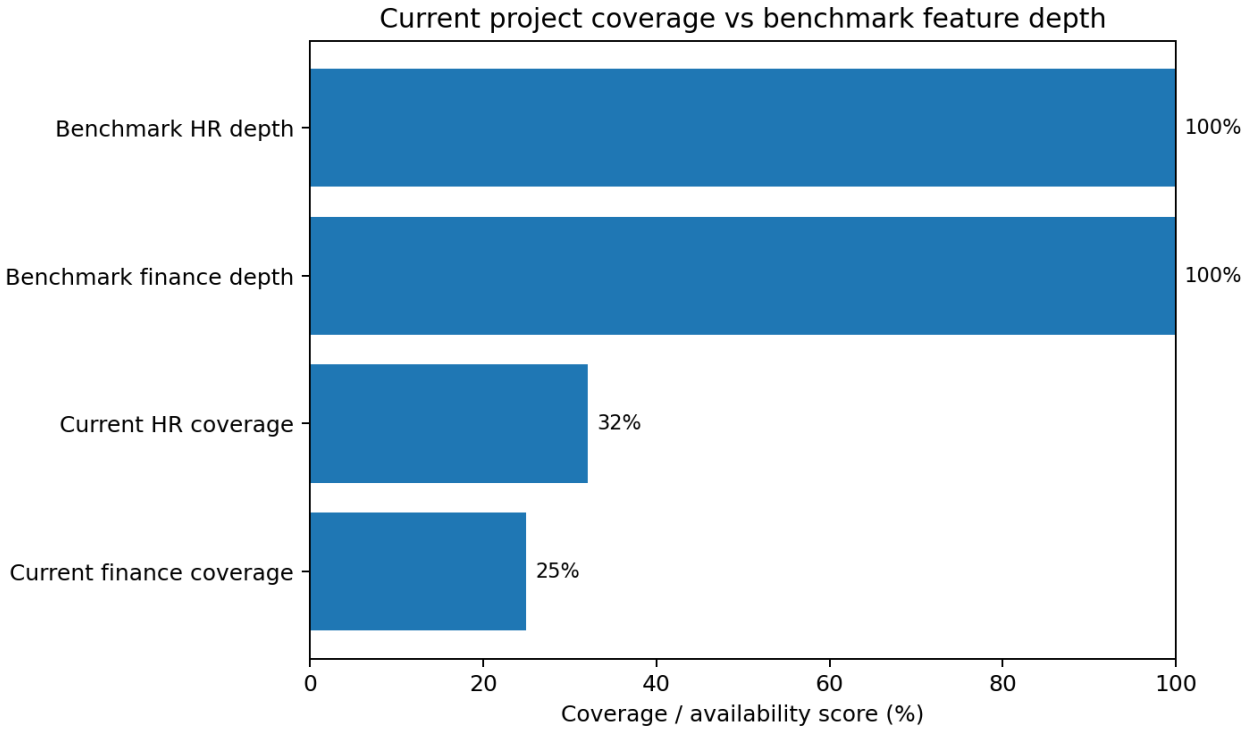


Figure 2. AIU current coverage vs benchmark feature depth.

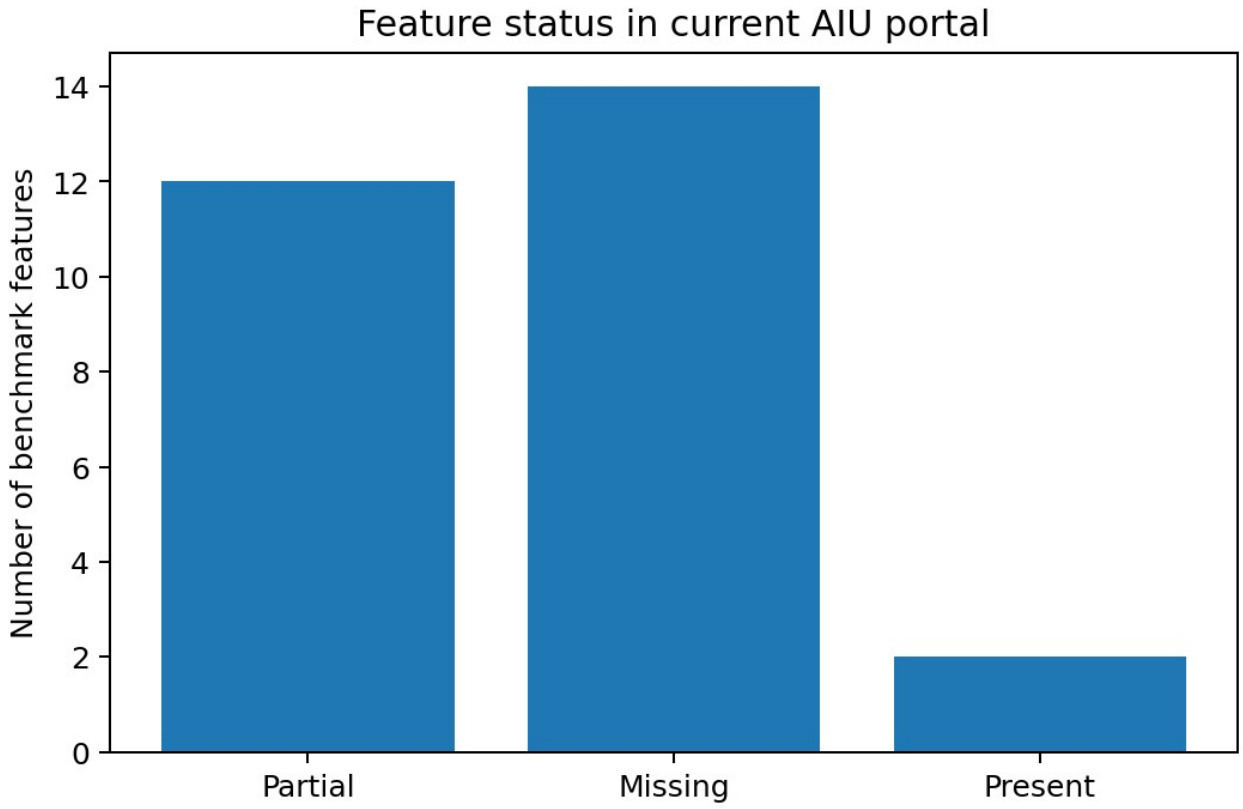


Figure 3. Count of present, partial and missing benchmark features.

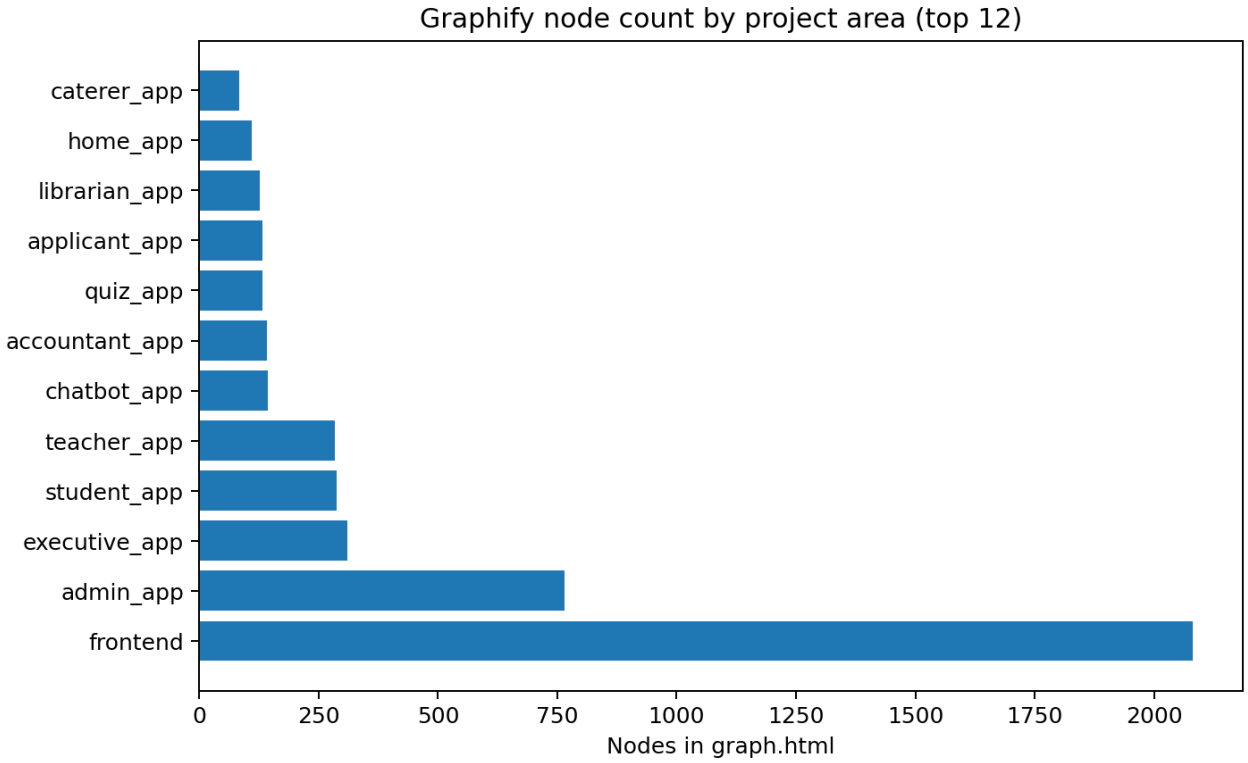


Figure 4. Graphify node count by project area.

3. Current project baseline from your documentation and Graphify output

Graphify report summary: 5665 nodes · 16050 edges · 345 communities (226 shown, 119 thin omitted). Graph HTML embedded summary: 4858 nodes · 12607 edges · 352 communities. The two uploaded graph artifacts appear to be from slightly different Graphify exports; the analysis treats the HTML graph as the inspectable node graph and the Markdown report as the broader corpus summary.

Domain	Implemented capability	Maturity
Public website & CMS	Home, about, admissions, scholarship, schools, centres, departments, programmes, units, services, student life, directories, news, events, announcements, official links, and dynamic pages.	Implemented
Identity & accounts	Username/password login, applicant registration, admin-created role accounts, profile picture updates, account recovery, role routing, and role-detail lifecycle.	Implemented
Admissions	Applicant profile, intake/programme options, draft application, scholarship data, documents, submission, status events, and staff review.	Implemented
Academic directory	Schools, centres, departments, units, programmes, courses, organizational scope, and public reuse of directory data.	Implemented
Semester administration	Calendar groups, registration phases, holidays, teaching/exam periods, planned/running semesters, notifications, and IELTS sessions.	Implemented

Course offerings & registration	Prerequisites, eligible schools, tutorial/lab groups, credit limits, capacities, registration periods, enrollment, executive manual registration, and history.	Implemented
Scheduling & calendar	Scoped executives create and maintain semester class schedules with lecturer/venue conflict checks. Teachers can adjust one class occurrence at a time, and students and teachers receive role-filtered timetable and combined academic-calendar views. Final exams and XLSX schedule export are also supported.	Implemented
Learning management	Course sections, folders, resources, course announcements, capacity-limited presentation slots, individual and group assignments, lab activities, upload rules, group invitations, leader-only final actions, rubrics, submissions, comments, feedback, and activity deadlines.	Implemented
Attendance	Teacher sessions, attendance codes, QR tokens, manual statuses, corrections, student submission logs, reports, and low-attendance notifications.	Implemented
Results & exams	Teacher result submissions, executive batches/review/publication, transcript records, GPA/CGPA views, coursework, semester PDFs, final exam schedules, venue allocations, and exam slips.	Implemented
Leave & student life	Student leave requests, teacher/dean approval chain, tracking history, emergency contacts, student affairs contacts, wardens, hostels, and hostel rooms.	Implemented
Venue booking	Student/teacher requests, admin approval/rejection, reasons, recurring-class conflicts, approved-booking conflicts, and final-exam integration.	Implemented
Elections	SRC and school-scoped elections, seat counts, candidate approval, exact ballot rules, result snapshots, publication, and notifications.	Implemented
Library	Book catalogue, physical copies, shelves, laptops, student borrow and acquisition requests, fulfillment, loans, returns, due reminders, and role restrictions.	Implemented
Canteen	Monthly allowance policy, eligibility, monthly bonuses, student accounts, caterer deductions, admin adjustments, and student balance history.	Implemented
Finance subscriptions	Accountant role, ToyyibPay/DuitNow-compatible gateway settings, gym and kitchen subscription plans, purchase checkout, callback activation, printable subscription cards, and role-based subscription access.	Implemented
ICT helpdesk	Student ticket submission, category/location, technician assignment, status updates, progress notes, resolution, and admin account management.	Implemented
Portal chatbot	Public AIU information answers, public directory email lookup,	Implemented

	privacy refusals, authenticated student/teacher timetable and assignment tools, optional Ollama-backed generation, and chatbot API tests.	
Notifications, reports & audit	Per-user notifications, unread summary/read actions, role fan-out, deep links, admin/executive exports, report history, structured change auditing, and scheduled assignment/library reminders.	Implemented

4. Existing finance and HR-adjacent capabilities in your project

Finance-adjacent features already present

Existing capability	How it helps expansion
Accountant role and routing	accountant_app + frontend accountant workspace; already supports a finance user persona.
Finance gateway settings	FinanceGatewayConfig and ToyyibPay/DuitNow-compatible gateway configuration already exist.
Subscription plan/payment attempts	SubscriptionPlan, Subscription and PaymentAttempt support gym/kitchen subscriptions and payment return/callback.
Student fines/charges	Fine model/service/views plus student/accountant fine pages, due/overdue handling, payment link creation and manual deduction from meal allowance.
Canteen allowance integration	Student canteen account, monthly allowance, deductions, adjustments and balance history can become a subledger source.
Notifications/reporting/audit	Shared notification system, report exports and audit logs can be reused for finance and HR approvals.

HR-adjacent features already present

Existing capability	How it helps expansion
Identity and role-detail models	Profile links users to Student, Teacher, Executive, Librarian, Caterer, Technician, Accountant and Admin role details.
Academic staff roles	Teacher/executive workflows exist, but they are academic roles, not full employee HR records.
Student leave and class attendance	Useful patterns for approval and attendance, but not yet staff leave or employee time tracking.
Surveys/evaluations	Can be adapted into staff performance review and appraisal cycles.
Admin account management	Can seed onboarding/offboarding workflows, but currently lacks HR appointment lifecycle.

5. Repository corpus summary

The comparison uses all 100 repositories from the previously generated GitHub list. The purpose is to extract reusable feature patterns, not to recommend copying code directly. License compatibility and code quality must be checked before reuse.

Cluster	Repository count	What this cluster contributes
Higher-ed finance	4	Deep university finance patterns: GL, labor distribution, grants, AP, travel, effort certification.
Other / platform	1	Workflow/API/platform examples that may help architecture but are not core finance/HR.
Education ERP / school management	37	Student administration plus fees, invoice, accounting, HR/payroll patterns in school/college context.
HR / payroll	44	Employee lifecycle, attendance, leave, payroll, deductions, payslips, self-service.
General ERP / accounting	14	Accounting, procurement, inventory/assets, reporting, vendor and workflow patterns.

6. Feature gap matrix: what other repos have but your project lacks

Interpretation: Present = implemented or directly documented. Partial = adjacent foundation exists but not a full module. Missing = no clear implementation in current documentation/Graphify evidence.

Feature/functionality	Current AIU status	Repos showing pattern	Example repos	Recommended action
Higher-ed financial system	Partial	3	kuali/kfs, kuali/kfs-db, CU-CommunityApps/cu-kfs	Extend existing model/service into full workflow. Accountant role, gateway configuration, subscriptions, payment attempts and fines exist, but not a full KFS-like finance system.
General ledger / chart of accounts	Missing	17	frappe/education, frappe/erpnext, okoyechuka/school-managment-software, ashikahmed-bd/School-management-and-accounting-software, LitGrey-Technologies/Student-Dynamics, odoo/odoo	Add as new module or submodule with role, audit and report support. No chart of accounts, journal entries, ledger posting, trial balance or double-entry finance model is documented.
Student fee structure	Missing	16	frappe/education, Akshat4112/SchoolErp, adams-okode/school-management-system, hrshadhin/school-management-system, fayazkhan121/SMS-system, okoyechuka/school-managment-software	Add as new module or submodule with role, audit and report support. Roadmap explicitly recommends adding tuition fee structures.
Student ledger / AR	Missing	1	nong04/human-resources-management	Add as new module or submodule with role, audit and report support. Roadmap explicitly recommends student ledger/outstanding balances/finance holds.
Invoices and receipts	Partial	11	hrshadhin/school-management-system, fayazkhan121/SMS-system, Pinnacle96/Edulab-Software, SPF-UiO/spbm, odoo/odoo, Dolibarr/dolibarr	Extend existing model/service into full workflow. Subscription receipt/card generation exists; student invoice and tuition receipt lifecycle is not documented.
Payment gateway / online payment	Present	6	hrshadhin/school-management-system, fayazkhan121/SMS-system, Pinnacle96/Edulab-Software, TareqMonwer/Django-School-Management, krishsharma5169/university-management-system, akaunting/akaunting	Keep and harden; integrate with audit/reporting. ToyyibPay/DuitNow-compatible gateway settings, subscriptions, callbacks and fine payment links are documented.
Payment reconciliation	Missing	1	artivisi/balaka	Add as new module or submodule with role, audit and report support. Roadmap recommends gateway reconciliation; no reconciliation workflow currently appears.
Scholarship / sponsorship offsets	Partial	2	kuali/kfs, kuali/kc	Extend existing model/service into full workflow. Admissions scholarship data exists; finance offsets/sponsorship

				posting are not implemented.
Budgeting / budget construction	Missing	3	hrshadhin/school-management-system, akaunting/akaunting, ever-co/ever-gauzy	Add as new module or submodule with role, audit and report support. No institutional budget, department budget or budget-control module is documented.
Procurement / purchasing / AP	Missing	36	kuali/kfs, kuali/kfs-db, CU-CommunityApps/cu-kfs, frappe/education, frappe/erpnext, frappe/hrms	Add as new module or submodule with role, audit and report support. Roadmap recommends procurement, suppliers and purchase approvals.
Asset management	Partial	6	pbse/College-Management-System---Ruby-On-Rails, odoo/odoo, Dolibarr/dolibarr, aureuserp/aureuserp, artivisi/balaka, benjdiasaad/ERP	Extend existing model/service into full workflow. Library laptops, venues/rooms and hostel directories exist, but no fixed-asset accounting lifecycle.
Travel / expense claims	Missing	8	kuali/kfs, Akshat4112/SchoolErp, hrshadhin/school-management-system, fayazkhan121/SMS-system, Pinnacle96/Edulab-Software, akaunting/akaunting	Add as new module or submodule with role, audit and report support. No expense claim/travel reimbursement workflow is documented.
Research grants administration	Missing	2	kuali/kfs, kuali/kc	Add as new module or submodule with role, audit and report support. Roadmap recommends research and ethics management; finance/grants are not implemented.
Financial reports	Partial	5	TAIJULAMAN/scholarstika--saas-development, Harishganth-0704/College-Management-System, Sairam-Radhakrishnan/PayFlow, ajithb236/PayrollManagementSystem, LucasMissiba/Ghost-RH	Extend existing model/service into full workflow. Reports and XLSX exports exist, but not accounting statements such as GL, trial balance, AR aging.
Employee master / HRIS	Partial	36	openeducat/openeducat_erp, frappe/hrms, hrshadhin/school-management-system, okoyechuka/school-managment-software, Kyalo-kekee/Kenya-SMS-Core, kngetich45/School-management-system	Extend existing model/service into full workflow. Teacher/executive/librarian/caterer/technician/accountant role details exist, but not a unified employee master/personnel file.
Recruitment / ATS	Missing	5	frappe/hrms, aureuserp/aureuserp, ever-co/ever-gauzy, apipulse/brbhr-frontend, preston-56/hr-management-system	Add as new module or submodule with role, audit and report support. No employee recruitment or job applicant tracking workflow is documented.
Onboarding / offboarding	Missing	1	frappe/hrms	Add as new module or submodule with role, audit and report support. Admin-created role accounts exist, but no HR appointment/onboarding/offboarding lifecycle.
Employee leave management	Partial	20	frappe/hrms, LitGrey-Technologies/Student-Dynamics, doptor/doptor,	Extend existing model/service into full workflow. Student leave

			phphrm/phphr, Gaurav241/OfficeOps-WPS, aryapatel23/Attendance-and-Payroll-Management	exists; staff/employee leave policies and approval flows are not documented.
Employee attendance / time tracking	Partial	29	navariltd/Junior-School, prajapatikaml/EduSec, Akshat4112/SchoolErp, pbse/College-Management-System---Ruby-On-Rails, kngetich45/School-management-system, adeshasur/Akuru-School-Managment-System	Extend existing model/service into full workflow. Class attendance exists for students; employee clock-in/timesheets are not documented.
Payroll engine	Missing	52	openeducat/ openeducat_erp, frappe/hrms, adams-okode/school-management-system, hrshadhin/school-management-system, fayazkhan121/SMS-system, okoyechuka/school-managment-software	Add as new module or submodule with role, audit and report support. No salary structure, payroll period, payroll run or employee compensation engine is documented.
Statutory deductions / tax	Missing	6	frappe/hrms, kngetich45/School-management-system, lizardcat/usiu-payroll-system, artivisi/balaka, KeneanDita/PayRoll-System, alfredang/ai-hrms	Add as new module or submodule with role, audit and report support. No payroll tax, statutory contributions, Zakat/EPF/SOCSO-style deductions or localization rules are documented.
Payslips / paycheck PDFs	Missing	5	Pinnacle96/Edulab-Software, lizardcat/usiu-payroll-system, aryapatel23/Attendance-and-Payroll-Management, thaokvu/PayFlow, Marwanthe0/Employee_ Management_Software	Add as new module or submodule with role, audit and report support. No employee payslip or payroll report PDF is documented.
Performance appraisal	Partial	7	openeducat/ openeducat_erp, frappe/hrms, meowmeow9567/College-Management-System, ArshnoorKaur21/Edutopia , Gaurav241/OfficeOps-WPS, digitalinnovations-tech/hr_management_system	Extend existing model/service into full workflow. Course evaluation and surveys exist; staff performance appraisal/review workflow is not implemented.
Training / learning	Partial	4	itsproali/university_core, jeffjeff2023/CollegeManagementSystem, thaokvu/PayFlow, KeneanDita/PayRoll-System	Extend existing model/service into full workflow. LMS exists for courses; employee training and professional development records are not documented.
Employee self-service	Partial	2	aryapatel23/Attendance-and-Payroll-Management, SafetyMP/HR-ERP	Extend existing model/service into full workflow. Role dashboards exist; HR self-service for staff profile, leave, payslips and claims is missing.
Organization positions / job roles	Partial	6	SPF-UiO/spbm, Harishganth-0704/College-Management-System, ArshnoorKaur21/Edutopia , hubleto/erp, thaokvu/PayFlow, preston-56/hr-management-system	Extend existing model/service into full workflow. Roles/scopes exist, but HR positions, grades, reporting managers and employment types are not documented.
Workflow approvals	Present	9	kuali/kc, kuali/rice, frappe/education,	Keep and harden; integrate with

			ashikahmed-bd/School-management-and-accounting-software, Harishganth-0704/College-Management-System, meowmeow9567/College-Management-System	audit/reporting. The portal already has approval patterns for admissions, leave, bookings, elections, results and reports.
Compliance / localization	Missing	4	frappe/hrms, odoo/odoo, yarsa/nepal-compliance, artivisi/balaka	Add as new module or submodule with role, audit and report support. No HR/finance localization framework is documented.

7. How to add Finance Management into the current project

Phase F1: Student billing and ledger foundation

- Add models: FeeCategory, FeeStructure, FeeItem, StudentCharge, StudentLedgerEntry, StudentBalanceSnapshot, StudentFinanceHold.
- Connect charges to AdmissionApplication, Enrollment, CourseRegistration, Hostel/Library/Canteen/Fine sources.
- Use immutable ledger entries; never update old financial rows except through reversal/adjustment entries.
- Expose APIs: /finance/fee-structures, /finance/student-ledgers, /finance/students/{id}/balance, /finance/holds.

Phase F2: Invoicing, receipts and payment allocation

- Add Invoice, InvoiceLine, Receipt, PaymentAllocation, RefundRequest and PaymentReconciliationBatch.
- Reuse existing PaymentAttempt and gateway callbacks, but allocate payments against invoice lines and ledger entries.
- Generate PDF invoices/receipts and store signed immutable copies with audit trail.
- Add accountant dashboard cards: billed, paid, outstanding, overdue, unallocated payments, failed callbacks.

Phase F3: Financial control and reporting

- Add chart of accounts and posting rules only after student ledger stabilizes.
- Add department/programme budgets, purchase requests, vendors, AP invoices and approvals.
- Add AR aging, trial balance, daily collection report, refund report, finance-hold report and scholarship-offset report.
- Export to XLSX/CSV and optionally integrate with ERPNext/Odoo/Kuali instead of rebuilding full GL.

Suggested finance model skeleton

```
class StudentCharge(models.Model):
    student = ForeignKey(Student)
    source_type = CharField()          # tuition, hostel, fine, library, canteen, manual
    source_id = PositiveIntegerField(null=True)
    amount = DecimalField()
    due_date = DateField()
    status = CharField(choices=[DRAFT, POSTED, WAIVED, REVERSED])

class StudentLedgerEntry(models.Model):
    student = ForeignKey(Student)
    charge = ForeignKey(StudentCharge, null=True)
    entry_type = CharField()           # debit, credit, payment, waiver, refund, reversal
    amount = DecimalField()
    balance_after = DecimalField()
    posted_by = ForeignKey(Profile)
    posted_at = DateTimeField(auto_now_add=True)

class PaymentAllocation(models.Model):
    payment_attempt = ForeignKey(PaymentAttempt)
    ledger_entry = ForeignKey(StudentLedgerEntry)
    amount = DecimalField()
```

8. How to add HR Management into the current project

Phase H1: Unified employee master

- Create hr_app and Employee model linked 1:1 to Profile, but do not remove Teacher/Executive/etc. role-detail models.
- Add EmploymentRecord, Position, DepartmentAssignment, SupervisorRelationship, ContractDocument and EmployeeDocument.
- Add staff status lifecycle: draft, active, probation, suspended, resigned, terminated, retired.
- Map existing role accounts into employee records with a migration command.

Phase H2: Staff leave and employee attendance

- Reuse student leave approval patterns but add LeaveType, LeavePolicy, LeaveEntitlement, LeaveRequest and LeaveApproval for employees.
- Add employee attendance/check-in with shift schedules, holiday calendars, late/absent rules and manual corrections.
- Expose self-service pages: My HR Profile, My Leave, My Attendance, My Documents.

Phase H3: Payroll

- Add SalaryStructure, SalaryComponent, EmployeeSalaryAssignment, PayrollPeriod, PayrollRun, PayrollLine, DeductionRule, Payslip.
- Support fixed salary, allowance, deduction, overtime, unpaid leave, advances and manual adjustment components.
- Lock payroll runs after approval; corrections should create adjustment lines in the next payroll period.
- Generate payslip PDFs, payroll journal export and bank-transfer file export.

Phase H4: Recruitment, onboarding and performance

- Add JobOpening, JobApplicant, Interview, Offer, OnboardingTask and OffboardingTask.
- Adapt existing survey/evaluation architecture into PerformanceCycle, AppraisalForm, ReviewSubmission and ReviewOutcome.
- Connect onboarding/offboarding to identity lifecycle: account creation, role provisioning, MFA, access revocation.

Suggested HR model skeleton

```
class Employee(models.Model):
    profile = OneToOneField(Profile)
    employee_no = CharField(unique=True)
    employment_status = CharField()
    joining_date = DateField()
    reporting_manager = ForeignKey('self', null=True)

class PayrollRun(models.Model):
    period = ForeignKey(PayrollPeriod)
    status = CharField(choices=[DRAFT, REVIEW, APPROVED, PAID, CANCELLED])
    approved_by = ForeignKey(Profile, null=True)

class Payslip(models.Model):
    payroll_run = ForeignKey(PayrollRun)
    employee = ForeignKey(Employee)
    gross_pay = DecimalField()
    total_deductions = DecimalField()
    net_pay = DecimalField()
    pdf_file = FileField(null=True)
```

9. Integration plan with your existing architecture

Existing project area	Integration action
Profile and role routing	Add Employee as a cross-role master; keep existing role details for academic permissions.
accountant_app	Keep subscriptions/fines; add finance_ledger.py, invoice_service.py,

	reconciliation_service.py and reporting endpoints.
admin_app	Add HR admin pages for positions, departments, employee creation, offboarding and role provisioning.
student_app	Expose student ledger, invoices, receipts, finance holds and payment history.
teacher/executive/librarian/caterer/technician roles	Map each role-detail record to Employee for HR, leave, payroll and documents.
notifications_package	Reuse for invoice issued, receipt paid, overdue, HR approval, payroll published, leave decision and document expiry alerts.
reports/audit	Reuse audit for every money/HR mutation; add export definitions for finance and payroll reports.
Celery beat/worker	Add scheduled jobs: invoice due reminders, reconciliation imports, payroll calculations, leave accruals, document expiry warnings.
Frontend	Create route groups /finance/*, /hr/* and self-service pages under existing role-zone navigation.

10. Prioritized backlog

Priority	Backlog item	Scope	Suggested size
P0	Student ledger core	StudentCharge, StudentLedgerEntry, invoice/receipt baseline, balance API, finance hold integration.	2-3 sprints
P0	Unified employee master	Employee linked to Profile, employment record, position, supervisor, documents, migration from existing staff roles.	2 sprints
P1	Invoice/payment allocation/reconciliation	Allocate existing gateway payments against invoices/fines/subscriptions; reconciliation batch; unallocated payment handling.	2-3 sprints
P1	Staff leave + attendance	Employee leave policy, entitlement, approvals, employee check-in/timesheet and correction workflow.	2-3 sprints
P1	Payroll MVP	Salary structure, payroll period/run, payslips, allowances/deductions, PDF and XLSX export.	3-4 sprints
P2	Budgets/procurement/AP	Purchase request, vendor, AP invoice, budget line, approval and reporting.	4+ sprints
P2	Recruitment/onboarding/ performance	Job openings, applicants, onboarding tasks, performance cycles and offboarding access revocation.	4+ sprints
P2	GL/accounting integration	Either build simple GL or integrate ERPNext/Odoo/Kuali export connector.	Depends on policy

11. Appendix A: all 100 repositories used in comparison

#	Repository	Category	Fit	Feature notes
1	kuali/kfs	Higher-ed finance	Strong	Kuali Financial System; comprehensive financial management system for higher education, including modules such as contracts & grants, labor distribution, purchasing/AP, travel/entertainment, and effort certification.
2	kuali/kfs-db	Higher-ed finance support	Strong	Database import/export, demo data, and bootstrap datasets for Kuali Financial System; useful when evaluating or setting up KFS.

3	CU-CommunityApps/cu-kfs	University finance customization	Strong	Cornell University customizations for Kuali Financial System; useful as a real university implementation/reference customization layer.
4	kuali/kc	Research finance/grants	Strong	Kuali Research/Kuali Coeus; research administration system for grants, sponsored projects, and faculty research administration workflows.
5	kuali/rice	Workflow/platform	Useful	Kuali Rice workflow/platform services used across Kuali higher-education administrative systems; useful if exploring legacy Kuali stack architecture.
6	openeducat/openeducat_erp	Education ERP	Strong	Education ERP for schools/colleges/universities. Search snippet highlights HR & Payroll Management for staff records, payroll, and performance reviews.
7	frappe/education	Education ERP	Strong	Frappe Education; open-source education/school management module built on ERPNext/Frappe, with administrative and academic workflows and ERPNext accounting integration for fees.
8	frappe/erpnext	ERP + education/finance	Strong	100% open-source ERP with accounting, operations, and education-domain support; useful as the base ERP when paired with Frappe Education and Frappe HR.
9	frappe/hrms	HR/payroll	Strong	Frappe HR / HRMS; open-source HR and payroll software covering employee management, onboarding, leave, payroll, taxation, recruitment, shifts, and performance.
10	fderyckel/ifitwala_ed	Education operating system	Strong	Ifitwala Ed; open-source Education Operational System for schools, colleges, and education groups with a unified permission model and analytics surface.
11	fderyckel/ifitwala_ed_v2	Education operating system	Strong	Ifitwala_Ed v2; described as a unified institutional operating system replacing fragmented SaaS tools with one data model and permissions system.
12	fderyckel/ifitwala_ed_V0	Education ERP variant	Useful	Earlier Ifitwala Ed / Frappe app based on ERPNext Education domain; useful for implementation history and ideas.
13	navariltd/Junior-School	Frappe Education extension	Useful	Frappe Education extension for scheduling, attendance, assessment, automated student lifecycle, and multi-school

				environments.
14	prajapatikaml/EduSec	College/university ERP	Strong	EduSec college management system; covers university functions including admission, administration, timetable, examination, HR, finance, library, transport, and attendance.
15	djoudi/EduSec2.0.0	College/university ERP	Useful	Fork/new version of EduSec 2.0.0; includes education-industry modules such as HR and finance according to README snippet.
16	Akshat4112/SchoolErp	School ERP	Strong	SchoolERP with student management, accounts, attendance, faculty management, HR management, fees, expenses, results, and marks.
17	pbse/College-Management-System---Ruby-On-Rails	College management	Strong	Ruby on Rails school/college management system with Student, Faculty/Non-Faculty, HR, Finance, messaging, hostel, inventory, online exam, and attendance modules.
18	adams-okode/school-management-system	School MIS	Strong	ALANA Open Source School MIS; includes finance management, HRM, payroll, student fees, accommodation, and library modules.
19	hrshadhin/school-management-system	School management	Strong	School system with account management, budget, account heads, student invoice, income/expense management, payroll, salary template, and employee salary payment.
20	fayazkhan121/SMS-system	School management fork/reference	Useful	School management system referencing CloudSchool/hrshadhin setup; includes student invoice, income/expense, payroll, salary templates, and salary payment in snippet.
21	okoyechuka/school-managment-software	School management	Strong	Open-source PHP school management software with fee module, financial accounting, double-entry accounting, HR module, employee payroll, pay-slip generation, and processing.
22	Kyalo-kekee/Kenya-SMS-Core	School MIS	Useful	K-SMS: Kenyan School Management Information System with student, class, staff, exam, HR, hostel, and fee management modules.
23	kngetich45/School-management-system	School management	Useful	School management system with fees, attendance, staff/student management, and easy payroll with allowances and deductions.
24	TAIJULAMAN/	Multi-tenant school SaaS	Strong	Scholarstika: multi-tenant

	scholarstika--saas-development			school management system with fees, transactions, earnings, payroll, financial reports, staff/student management, and branch payroll administration.
25	Pinnacle96/Edulab-Software	School SaaS	Strong	Edulab SaaS multi-tenant school management system with finance: fee management, invoicing, online payments, expense tracking; HR & Payroll: staff management, payroll, payslips.
26	Dantechdevs/CBC-school	School management	Useful	ElimuMS / CBC school management system for Kenyan schools; includes fees and finance management and broader operational modules.
27	musaibasrar/Curium	School management	Useful	Curium comprehensive school management software; includes fee collection and mess/payroll according to README snippet.
28	hk-kumawat/College-Management-System	College management	Useful	Digital solution for college operations, student/teacher management, fee tracking, and record keeping.
29	TareqMonwer/Django-School-Management	School management	Useful	Django school management system; snippet highlights payments, e-admissions, and results management.
30	ashikahmed-bd/School-management-and-accounting-software	School accounting	Strong	School Management and Accounting Software; useful for education finance/accounting workflows and local customization ideas.
31	LitGrey-Technologies/Student-Dynamics	Education ERP	Useful	Student Dynamics ERP for educational institutes; topics include HR, HRMS, leave management, accounting software, library, admissions, SMS, and school management.
32	aisuresh/School-Management-System	School management	Useful	SaaS-based school management system for school-related activities and process automation.
33	adeshasur/Akuru-School-Managment-System	Education ERP	Useful	Akuru school ERP for Sri Lankan schools; handles core operations including student management, attendance, exam grading, and finance management.
34	SPF-UiO/spbm	University student personnel/payroll	Useful	Student Personnel Booking Manager for University of Oslo student pub personnel organization; topics include Django, invoicing, booking, student management, and payroll.
35	Emalsha/hris	University HRIS	Useful	Human Resource Information System for University of Colombo School of Computing by

				team Helix.
36	lizardcat/usiu-payroll-system	University payroll	Useful	USIU-Africa Payroll System for a hypothetical university in Kenya; salary computation, statutory deductions, and payslip generation.
37	Marwanthe0/Payroll_Management_Software	University payroll	Useful	FastAPI + Streamlit payroll management software for a university according to GitHub topic snippet.
38	krishsharma5169/university-management-system	University management	Limited	Python university management system for student registration, course enrollment, attendance, and payments.
39	itsproali/university_core	University core service	Limited	University Management Core Service for academic operations such as courses, semesters, registration, and student management.
40	jeffjeff2023/CollegeManagementSystem	College management	Limited	College management system for automating student records, faculty details, courses, and related operations.
41	bonitoflakez/college-student-management-system	College management	Limited	React/TypeScript/Node/PostgreSQL web app for college operations and student experience management.
42	themohitshaw/KMBB-CMS	College CMS	Limited	Flask/MySQL college management system for students, faculty, notices, and academic activities.
43	shshwtsrkr/College-Management-System	College management	Limited	Full-stack college management website with PHP/MySQL backend and HTML/Bootstrap/JavaScript frontend.
44	Harishganth-0704/College-Management-System	College management	Limited	MERN college management system with admin, faculty, and student roles, attendance tracking, notes, notifications, and reports.
45	meowmeow9567/College-Management-System	College/school management	Limited	MERN school/college management system for students, faculty, classes, attendance, performance, marks, and communication.
46	ArshnoorKaur21/Edutopia	School management	Limited	MERN school management app for class organization, student/faculty records, attendance, assessment, and feedback.
47	AmitXShukla/Online-School-Management-App-Angular-Firebase	School management	Limited	Angular/Firebase online school management app for students, records, grades, fees, attendance, staff, and notifications.
48	odoo/odoo	General ERP	Strong	Odoo Community; web-based open-source business apps including billing/accounting, project management, HR, inventory, manufacturing, marketing, and more.

49	OCA/hr	Odoo HR addons	Strong	Odoo Community Association HR addons; useful for extending Odoo HR functionality in an open-source/community stack.
50	OCA/payroll	Odoo payroll addons	Strong	Odoo payroll management modules from OCA; useful for payroll on top of Odoo Community.
51	yarsa/nepal-compliance	ERPNext compliance	Useful	ERPNext compliance app for Nepal covering HR, Payroll, and Accounting modules; useful as an ERPNext localization/compliance pattern.
52	Dolibarr/dolibarr	General ERP/CRM	Strong	Dolibarr ERP/CRM for contacts, suppliers, invoices, orders, stocks, agenda, accounting, and other business/foundation activities.
53	aureuserp/aureuserp	General ERP	Strong	Free and open-source ERP platform with sales, accounting, maintenance, invoice, manufacturing, recruitment, inventory, HRM, warehouse, PMS, and purchasing modules.
54	akaunting/akaunting	Accounting/finance	Strong	Online accounting software covering finance, invoices, payments, payroll, billing, budgets, expenses, and bookkeeping.
55	idurar/idurar-erp-crm	ERP/CRM accounting	Useful	Free open-source ERP/CRM software with accounting and invoicing built with Node.js and React.
56	metasfresh/metasfresh	General ERP	Useful	Open-source ERP described as fast, flexible, and free; includes business, accounting, CRM, manufacturing, and PostgreSQL-based operations.
57	apache/ofbiz-framework	General ERP framework	Strong	Apache OFBiz ERP framework with broad business-process automation; official snippet says it houses libraries, entities, services, and features for business operations.
58	apache/ofbiz-plugins	OFBiz plugins	Useful	Complementary plugins for Apache OFBiz, adding enterprise/e-commerce/business features to the framework.
59	ever-co/ever-gauzy	ERP/CRM/HRM/ATS	Strong	Ever Gauzy open business management platform covering ERP, CRM, HRM, ATS, project management, employees, finance, invoices, expenses, time tracking, and budgets.
60	wp-erp/wp-erp	WordPress ERP	Useful	Open-source ERP solution for WordPress with accounting, CRM, and HR topics/modules.
61	notrinos/NotrinosERP	ERP/accounting/HR	Strong	Web-based

				ERP/accounting system in PHP/MySQL including CRM, sales, purchasing, warehousing, manufacturing, payroll, and human resource modules.
62	minthcm/minthcm	HCM	Useful	Open-source HCM system for managing HR processes with full data ownership and no vendor lock-in.
63	hubleto/erp	ERP/CRM	Useful	Hubleto ERP/CRM core with free-to-use apps for managing organizations; customizable and fully open source according to topic snippet.
64	iota-uz/iota-sdk	Modular ERP	Useful	Open-source modular ERP alternative to SAP/Oracle/Odoo; topic snippet mentions ERP finance.
65	artivisi/balaka	Accounting + payroll	Useful	Open-source accounting application for Indonesian SMEs with tax compliance, payroll, inventory, bank reconciliation, and invoicing.
66	benjdiasaad/ERP	ERP	Useful	Multi-company ERP with sales, purchasing, inventory, finance, HR, CRM, projects, events, permissions, and tenant isolation.
67	ItzKssBerc/comproller	ERP	Useful	Modern Laravel ERP with finance, HR, attendance monitoring/tracking, payroll-management topics, and Filament UI.
68	doptor/doptor	Accounting/HR/payroll/ college	Useful	Doptor platform mentioned in Laravel OSS lists as including accounting, HR, payroll management, and college management modules.
69	phphrm/phphr	HRMS/payroll	Strong	PHPHR Open-Source Edition; HR management system with employee, attendance, leave, and payroll modules.
70	Devnawjesh/hr-payroll	HR/payroll	Strong	Modern responsive HRM built with PHP/Laravel for office and employee management.
71	MusheAbdulHakim/ Smarthr-hr-payroll-project-employee-management-System	HR/payroll	Useful	Smart HR management system with PHP and MySQL; useful for employee management/payroll workflows.
72	ahmedsaadawi13/ splash-hr	HR/payroll	Strong	Complete HR and Payroll management system in PHP/MySQL with multi-tenancy, HR features, and REST API integration.
73	Gaurav241/OfficeOps-WPS	Workforce/HR	Useful	Django Workforce Productivity Suite for HR and office management, attendance, feedback, leave, and employee operations.
74	aryapatel23/	Attendance/payroll	Useful	Full-stack attendance and

	Attendance-and-Payroll-Management			payroll management with HR/admin and employee dashboards, salary slip generation, leave, holidays, and payroll processing.
75	shashikiran-dev/Payroll-Management-System	Payroll	Useful	Scalable payroll management with microservice architecture, attendance, leave processing, WFH tracking, analytics, and chatbot integration.
76	thaokvu/PayFlow	Payroll	Useful	Secure Flask payroll management system with paycheck generation, roles, financial data, and PDF paychecks.
77	Sairam-Radhakrishnan/PayFlow	Payroll	Useful	Employee Payroll Management System for employee records, salary calculations, and reports for full-time/part-time employees.
78	Duisternis/Payroll-Management-System	Payroll	Useful	Python payroll management application for employee payroll and salary processing; MIT license shown in search snippet.
79	emmanpbarrameda/Employee-Payroll-Management-System-Java-GUI	Payroll	Useful	Java Swing Employee Payroll Management System with installer, desktop app, and MySQL database.
80	ajithb236/PayrollManagementSystem	Payroll	Useful	Django web-based payroll system for compensation, attendance tracking, leave management, and financial reporting.
81	KeneanDita/PayRoll-System	Payroll	Limited	Java console payroll management system for salary calculations, tax deductions, and employee records.
82	RJGATON007/hr-payroll	HR/payroll	Useful	Human Resource Management System in PHP CodeIgniter and Laragon; payroll/HRMS focus.
83	Ranazia943/multi-company-payroll-attendance-system	HR/payroll	Useful	Multi-company HR & Payroll system with fingerprint attendance, leave, and salary management.
84	MiraiNetwork/hr-platform	HR/payroll	Useful	Open-source HR management platform covering employee, attendance, leave, shift, accounting, payroll, and multi-tenant SaaS topics.
85	SafetyMP/HR-ERP	HR/payroll SaaS scaffold	Useful	Open-source reference for multi-tenant HR SaaS with payroll kernel, employee self-service/manager flows, and governance harness.
86	nong04/human-resources-management	Odoo HR modules	Useful	Odoo 17 HR management suite with core HR, leave balances, systray attendance, and automated payroll.
87	4ban-university/HRMS-MicroServices	HRMS microservices	Useful	Software project management HRMS

				microservices repository with payroll folder/topic; Java/Spring microservice reference.
88	lagunadevs/hris	HRIS/payroll	Useful	SEA HR Systems: payroll, leave management, and time attendance management.
89	apipulse/brbhr-frontend	HRMS frontend	Useful	Frontend for brbhr HRMS built with React/Chakra UI; aims at production-grade HRMS with hiring, attendance, payroll, leave, and enterprise HR topics.
90	digitalinnovations-tech/hr_management_system	HR management	Useful	HR management system for employees, payroll, attendance, leave requests, performance tracking, and secure admin dashboard.
91	Trevor-Kayeyia-Madara/Vegro-HR-Management-Platform	HR platform	Useful	Modern HR platform for employee management, payroll, attendance, and access control built with PHP/Laravel concepts.
92	dev-javid/bit-hrms	HRMS	Useful	.NET + React HRMS with employee management, attendance tracking, leave management, and salary processing.
93	preston-56/hr-management-system	HR management	Useful	Next.js HR management system for employees, recruitment, leave tracking, performance reviews, and organizational workflows.
94	alexisbriandev/php-attendance-management-system	Attendance/leave	Useful	PHP Attendance Management System for HR attendance lifecycle; supports leave-management topic.
95	LucasMissiba/Ghost-RH	HR support	Useful	HR support system with QR-code time tracking, vacation management, document upload, reporting, and dashboard.
96	Xilantra/coralhr	HR/payroll	Limited	HRMS/payroll-related repository with broad HR software keywords; evaluate actual code before use.
97	alfredang/ai-hrms	HRMS	Useful	AI-powered HRMS with employee management, leave tracking, Singapore CPF payroll, expense claims, and chatbot.
98	Marwanthe0/Employee_Management_Software	Employee/payroll	Useful	Employee management software with add/delete/update employees, pay amount adjustment, and monthly payslip printing.
99	hassanmahfuj/HR-Glance	HRMS	Useful	Java HR management system for personal data, attendance, leaves, and payroll.
100	SupportComeos/HR-API	HR API	Limited	Spring Boot API for an HR SaaS solution.

12. Sources and evidence used

- Uploaded project documentation: documentation.html (AIU Unified Portal — Project Documentation).

- Uploaded Graphify files: GRAPH_REPORT(1).md and graph.html.
- Previously generated repo list: open_source_university_finance_hr_github_repos.docx / repos.json containing 100 GitHub repositories.
- Representative GitHub/official project pages checked for major reference projects: Kuali KFS, Frappe Education, ERPNext, Frappe HRMS, OpenEduCat, Odoo/OCA, Dolibarr, OrangeHRM, Axelor, Apache OFBiz and related repos.